Pengxiang (Eric) Wang

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EDUCATION

University of Toronto Sep 2023 – Apr 2028 (Expected)

B.Sc. Computer Science (Specialist) GPA: 3.78 / 4.00

Coursework: CSC413 Neural Networks & Deep Learning, CSC415 Reinforcement Learning, CSC384 Intro to Artificial Intelligence, CSC311 Intro to Machine Learning, CSC373 Algorithm Design & Analysis, CSC363 Computational Complexity & Computability, CSC369 Operating Systems

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, SQL

AI / ML: PyTorch, HuggingFace Transformers, QLoRA, PEFT, model distillation, RAG

Evaluation: ablation studies, Exact Match, SQL execution checks, pytest, ruff, MOS, Whisper WER

LLM Infra: FastAPI, vLLM, Spring AI, LangChain4j, PgVector, Ollama, MCP, SSE streaming

Engineering: Spring Boot, PostgreSQL, Redis, Docker, uv, React, Vue

PROJECTS

Qwen2.5 QLoRA Text-to-SQL Benchmark

PyTorch · HuggingFace Transformers · PEFT · QLoRA · vLLM · FastAPI · sqlglot | GitHub | Hugging Face

- Built a reproducible QLoRA pipeline for `Qwen/Qwen2.5-1.5B-Instruct` on schema-conditioned Text-to-SQL, covering dataset conversion, YAML configs, training, evaluation, and reports.
- Ran LoRA rank ablations (`r=8/16/32`) on a 5k-train / 500-eval split; the best adapter improved Exact Match from 0.044 to 0.712 while keeping SQL parse validity at 0.990.
- Added SQL quality gates with Exact Match, `sqlglot` parse checks, error taxonomy, and a 30-case SQLite execution check with 1.000 parse validity and 0.600 execution accuracy.
- Shipped a local FastAPI demo and single-request serving benchmark comparing Transformers and vLLM, with tests, lint, technical reports, and a packaged Hugging Face LoRA adapter.

Circuit Agent — Full-Stack LLM Agent Platform

Spring AI 1.0 · Java 21 · PgVector · SSE · MCP · Docker | GitHub

- Built a dual-mode AI prototype: a domain RAG chatbot and an autonomous ReAct agent with 7 tools, manual context management, and session-scoped execution under the Java/Spring ecosystem.
- Reduced perceived response latency by switching from buffered to SSE streaming, cutting time-to-first-token from ∼1.2s to ∼0.5s under a 20-step deterministic execution budget.
- Implemented PgVector retrieval, pre-retrieval query rewriting, manual tool dispatch, a 20-step execution budget, and a `TerminateTool` for explicit loop termination.

NeuTTS-Express — Expressive TTS via Synthetic Distillation

PyTorch · Knowledge Distillation · HuggingFace Transformers · Neural Audio Codec | GitHub | Paper

- Designed the synthetic distillation setup for expressive TTS: a 3B teacher generated emotion-labeled audio to train a 0.5B student, reducing dependence on costly human-annotated emotional speech datasets.
- Built the evaluation pipeline measuring Whisper WER, real-time factor, and batch stability; the student runs 2.16× faster than real time with ∼4× lower WER variance (std 2.1% vs. 7.9%).
- Ran an 11-participant MOS study across 7 emotions; the student outperformed its 6× larger teacher on naturalness (3.99 vs. 3.17) and emotion match (3.91 vs. 2.82).

Shield-Guided Policy Stabilization for Safe Reinforcement Learning

JAX · DreamerV3 · AMBS · Python | GitHub | Paper

- Modified an AMBS / DreamerV3 shielded RL codebase by adding a threshold-based cost penalty and KL-regularized actor updates to reduce boundary oscillation under shielded training.
- Ran an 8-group ablation (baseline / cost-penalty / KL / full) across 3 safety labels in Atari Seaquest; the full method cut total safety violations from 272 to 112 over 45k steps while keeping task reward competitive.